

SKILLBRIDGE AI

AI-Powered Career Roadmap Generator

Internship Project Report

Submitted

By:

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Technology Used: Python, Flask, HTML, CSS, JavaScript, Google Gemini AI

SDG Supported: Sustainable Development Goal 8 – Decent Work & Economic Growth

1. Introduction

SkillBridge AI is an AI-powered web application developed to provide personalized career guidance for students. The application analyzes a student's career goals, current skills, and educational background using Google Gemini AI. Based on this analysis, it generates a customized career roadmap, identifies skill gaps, recommends projects, and provides interview and resume guidance.

The main objective of the project is to simplify career planning through Artificial Intelligence and help students become industry-ready.

2. Problem Statement

Many students struggle to identify the right career path due to:

- Lack of personalized career guidance.
- Difficulty identifying required technical skills.
- No structured learning roadmap.
- Limited awareness of industry requirements.
- Poor preparation for internships and placements.

Traditional career guidance methods provide generic advice and do not consider the individual skills or career goals of students.

3. Objectives

The objectives of SkillBridge AI are:

- To provide AI-powered career guidance.
- To analyze a student's current skills.
- To identify missing skills required for a chosen career.
- To generate a personalized learning roadmap.
- To recommend suitable projects.

- To provide interview preparation tips.
- To improve career readiness.
- To support Sustainable Development Goal 8.

4. Existing System

Existing career guidance systems mainly provide general information and static recommendations. Most platforms do not analyze an individual's profile or generate personalized career paths. Students often receive the same advice regardless of their educational background or current skills.

Limitations of Existing Systems

- Generic career advice
- No AI-based analysis
- No personalized roadmap
- Limited career planning assistance
- Lack of skill gap identification

5. Proposed System

SkillBridge AI is a web-based application that uses Artificial Intelligence to generate customized career guidance.

The user enters:

- Name
- Career Goal
- Current Skills
- Educational Qualification

The system processes the information using Google Gemini AI and generates:

- Career Analysis
- Skill Gap Report
- Learning Roadmap
- Recommended Projects
- Interview Preparation Tips
- Resume Suggestions

6. Methodology

The working process of SkillBridge AI consists of the following steps:

1. User opens the website.
2. User enters career information.
3. Flask backend receives the data.
4. Google Gemini AI analyzes the profile.
5. AI generates a personalized career roadmap.
6. The results are displayed on the website.

7. Technology Stack

Frontend

- HTML5
- CSS3
- JavaScript

Backend

- Python
- Flask Framework

Artificial Intelligence

- Google Gemini API

Development Tools

- Visual Studio Code
- Git
- GitHub

8. Key Features

- AI-powered Career Analysis
- Skill Gap Detection
- Personalized Career Roadmap
- Weekly Learning Plan
- Project Recommendations
- Interview Preparation

- Resume Improvement Tips
- Responsive User Interface
- Modern Web Design

9. Learning Objectives

This project helps students and developers understand:

- Flask Web Development
- Python Programming
- Google Gemini AI Integration
- Prompt Engineering
- HTML, CSS and JavaScript
- API Integration
- Career Recommendation Systems
- Git and GitHub Workflow

10. SDG Mapping

Sustainable Development Goal 8

Decent Work and Economic Growth

SkillBridge AI supports SDG 8 by:

- Improving employability.
- Encouraging continuous learning.
- Helping students identify skill gaps.
- Preparing students for internships.
- Supporting career development.
- Promoting equal access to career guidance.

11. Advantages

- Personalized career guidance.
- AI-generated recommendations.
- Easy-to-use interface.
- Fast response generation.
- Structured learning roadmap.

- Supports informed career decisions.
- Encourages self-learning.

12. Limitations

- Requires an internet connection.
- Depends on Google Gemini AI availability.
- AI recommendations may require human verification.
- Limited to text-based career guidance.

13. Future Scope

Future improvements may include:

- User Login System
- Progress Tracking Dashboard
- Resume Analyzer
- AI Mock Interview
- PDF Career Report Download
- Course Recommendation Engine
- Mobile Application
- Career Analytics Dashboard
- Multi-language Support

14. Conclusion

SkillBridge AI successfully demonstrates how Artificial Intelligence can be used to provide personalized career guidance. The system analyzes student profiles, identifies skill gaps, and generates structured learning roadmaps that help users prepare for internships and future careers.

The project contributes towards United Nations Sustainable Development Goal 8 – Decent Work and Economic Growth by promoting skill development, employability, and career readiness.

15. Project Repository

GitHub Repository

Repository Name: SkillBridge-AI

Project Link:

<https://github.com/tanvi251105/SkillBridge-AI>

Repository Contents

- Source Code
- HTML, CSS & JavaScript Files
- Flask Backend
- Google Gemini AI Integration
- README Documentation
- Requirements File
- Project Structure

16. References

1. Python Official Documentation
<https://www.python.org/>
2. Flask Documentation
<https://flask.palletsprojects.com/>
3. Google Gemini API Documentation
<https://ai.google.dev/>
4. GitHub Documentation
<https://docs.github.com/>
5. United Nations Sustainable Development Goal 8
<https://sdgs.un.org/goals/goal8>
6. Visual Studio Code Documentation
<https://code.visualstudio.com/docs>

17. Project Folder Structure

SkillBridge-AI

